

Theorem. (A conjecture of OEIS A051293, 2002) Let a_n denote the number of nonempty subsets of $\{1, 2, 3, \dots, n\}$ whose elements have an integer average. Then

$$a_n = \frac{2^{n+1}}{n} \left(1 + \frac{1}{n} + \frac{3}{n^2} + \frac{13}{n^3} + \frac{75}{n^4} + \frac{541}{n^5} + o\left(\frac{1}{n^5}\right) \right)$$

Proof. A subset $S \subseteq \{1, 2, \dots, n\}$ has an integer average if and only if the sum of its elements is divisible by its cardinality $|S|$. Let $N(n, k)$ denote the number of subsets of size k whose sum is divisible by k . We can express the total number of such subsets as $a_n = \sum_{k=1}^n N(n, k)$. To isolate the condition that k divides the sum of the elements in S , we utilize the orthogonality of roots of unity. Let $\omega_{k,j} = \exp\left(\frac{2\pi i j}{k}\right)$. By the standard orthogonality relation, the sum $\frac{1}{k} \sum_{j=0}^{k-1} \omega_{k,j}^N$ equals 1 if k divides N , and 0 otherwise. Thus, the indicator function for divisibility by k is $\frac{1}{k} \sum_{j=0}^{k-1} \omega_{k,j}^{\sum_{s \in S} s}$. Summing this over all $\binom{n}{k}$ subsets of size k yields

$$N(n, k) = \frac{1}{k} \sum_{j=0}^{k-1} \sum_{|S|=k} \omega_{k,j}^{\sum_{s \in S} s}$$

We isolate the principal term corresponding to $j = 0$, which evaluates trivially to $\frac{1}{k} \binom{n}{k}$. Let the remainder term be $R_{n,k}$, defined as

$$R_{n,k} = \frac{1}{k} \sum_{j=1}^{k-1} \sum_{|S|=k} \omega_{k,j}^{\sum_{s \in S} s}$$

Thus, $N(n, k) = \frac{1}{k} \binom{n}{k} + R_{n,k}$. Summing over all possible subset sizes k , we obtain

$$a_n = \sum_{k=1}^n \frac{1}{k} \binom{n}{k} + \sum_{k=1}^n R_{n,k}$$

Using the well-known identity $\sum_{k=1}^n \frac{1}{k} \binom{n}{k} = \sum_{j=1}^n \frac{2^j - 1}{j}$, we separate this into two sums: $S_n = \sum_{j=1}^n \frac{2^j}{j}$ and the harmonic number $H_n = \sum_{j=1}^n \frac{1}{j}$. This yields our fundamental decomposition

$$a_n = S_n - H_n + \sum_{k=1}^n R_{n,k}$$

Next, we establish that the sum of the remainders $\sum_{k=1}^n R_{n,k}$ is asymptotically negligible. Consider the inner sum $\sum_{|S|=k} \omega_{k,j}^{\sum_{s \in S} s}$. This sum is exactly the coefficient of z^k in the polynomial $P_{n,k,j}(z) = \prod_{m=1}^n (1 + z \omega_{k,j}^m)$. By standard coefficient bounds, the magnitude of the k -th coefficient of a polynomial is bounded by its maximum modulus on the unit circle $|z| = 1$:

$$\left| \sum_{|S|=k} \omega_{k,j}^{\sum_{s \in S} s} \right| \leq \max_{|z|=1} \prod_{m=1}^n |1 + z \omega_{k,j}^m|$$

For a fixed k and $j \in \{1, \dots, k-1\}$, the sequence of roots $\omega_{k,j}^m$ is periodic with period $L = \frac{k}{\gcd(j,k)} \geq 2$. The product over a full period L can be evaluated algebraically: the roots of $X^L - 1$ dictate that $\prod_{m=A}^{A+L-1} (1 + z\omega_{k,j}^m) = 1 - (-z)^L$. For any z on the unit circle, the magnitude $|1 - (-z)^L|$ is strictly bounded by 2. When taking the product over all n terms, we group them into $\lfloor n/L \rfloor$ full periods, leaving $n \bmod L$ residual terms. For $|z| = 1$, each full period contributes at most a factor of 2, and by the triangle inequality $|1 + z\omega_{k,j}^m| \leq 2$, each residual term also contributes at most a factor of 2. Therefore, the total product is bounded by $2^{\lfloor n/L \rfloor + (n \bmod L)}$. Because $L \geq 2$, we have $\lfloor n/L \rfloor + (n \bmod L) \leq \frac{n}{2} + 1$. Thus, for any z on the unit circle, we have

$$|P_{n,k,j}(z)| \leq 2^{\frac{n}{2}+1} = 2 \cdot 2^{n/2}$$

Consequently, the magnitude of the coefficient is bounded by $2 \cdot 2^{n/2}$. Applying this to our remainder definition yields $|R_{n,k}| \leq \frac{k-1}{k} 2 \cdot 2^{n/2} \leq 2 \cdot 2^{n/2}$. Summing over $k \in \{1, \dots, n\}$ gives the strict bound

$$\left| \sum_{k=1}^n R_{n,k} \right| \leq 2n2^{n/2}$$

We must now evaluate the asymptotics of $S_n = \sum_{j=1}^n \frac{2^j}{j}$. We propose the asymptotic approximation f_n , defined as

$$f_n = \frac{2^{n+1}}{n} \left(1 + \frac{1}{n} + \frac{3}{n^2} + \frac{13}{n^3} + \frac{75}{n^4} + \frac{541}{n^5} \right) = \frac{2^{n+1}P(n)}{n^6}$$

where $P(n) = n^5 + n^4 + 3n^3 + 13n^2 + 75n + 541$. To demonstrate that f_n is a highly accurate approximation of S_n , we analyze the discrepancy in their consecutive differences. Let E_j be the discrete derivative error given by

$$E_j = f_j - f_{j-1} - \frac{2^j}{j}$$

By telescoping summation, $S_n - f_n = S_1 - f_1 - \sum_{j=2}^n E_j$. To bound E_j , we expand $f_j - f_{j-1}$ algebraically to obtain

$$f_j - f_{j-1} - \frac{2^j}{j} = \frac{2^j [2P(j)(j-1)^6 - P(j-1)j^6 - j^5(j-1)^6]}{j^6(j-1)^6}$$

Let $A(x) = 2P(x)(x-1)^6 - P(x-1)x^6 - x^5(x-1)^6$. Expanding $A(x)$ leaves a polynomial of strictly degree 5:

$$A(x) = -4683x^5 + 13586x^4 - 19540x^3 + 15356x^2 - 6342x + 1082$$

In particular, it is easy to show that $|A(j)| \leq 100000j^5$ for $j \geq 2$. Substituting this back into the error term yields

$$|E_j| \leq \frac{100000 \cdot 2^j}{j^7}$$

Summing these errors up to n , the sum is dominated by its largest terms, giving $\sum_{j=2}^n |E_j| = \mathcal{O}\left(\frac{2^n}{n^7}\right)$. Recall our decomposition $a_n = f_n + (S_n - f_n) - H_n + \sum_{k=1}^n R_{n,k}$. To prove the theorem, it suffices to show that the combined error terms are $o\left(\frac{2^{n+1}}{n^6}\right)$:

1. $S_n - f_n = \mathcal{O}\left(\frac{2^n}{n^7}\right) = o\left(\frac{2^{n+1}}{n^6}\right)$.
2. $H_n \sim \ln n = o\left(\frac{2^{n+1}}{n^6}\right)$.
3. $\sum R_{n,k} \leq 2n2^{n/2} = o\left(\frac{2^{n+1}}{n^6}\right)$.

Since all residual terms are bounded by $o\left(\frac{2^{n+1}}{n^6}\right)$, we conclude that

$$a_n = f_n + o\left(\frac{2^{n+1}}{n^6}\right) = \frac{2^{n+1}}{n} \left(1 + \frac{1}{n} + \frac{3}{n^2} + \frac{13}{n^3} + \frac{75}{n^4} + \frac{541}{n^5} + o\left(\frac{1}{n^5}\right)\right)$$

This establishes the conjectured asymptotic expansion. □